Chapter 13

Vectors I

Contents

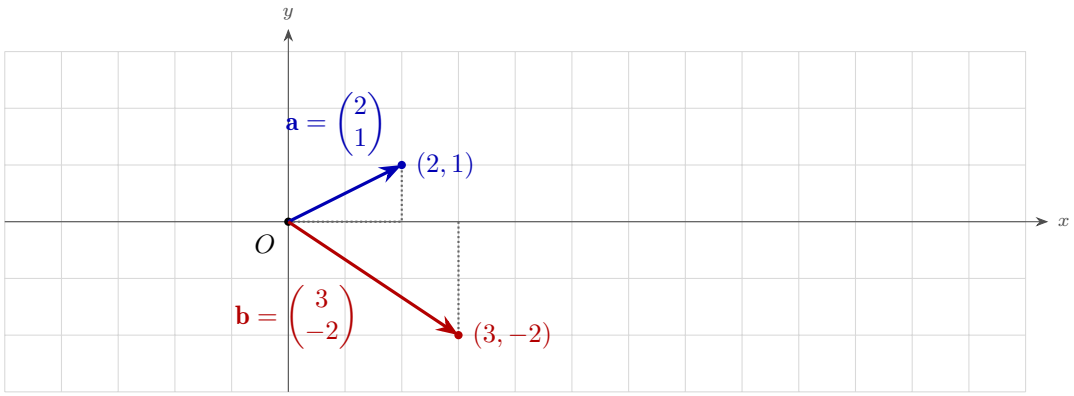
13.1 Vectors in Two Dimensions 2
13.2 Vectors in Three Dimensions 4
13.3 Basic Terminology and Notation 6
13.3.1 Ratio Theorem 8
13.4 Scalar Product 9

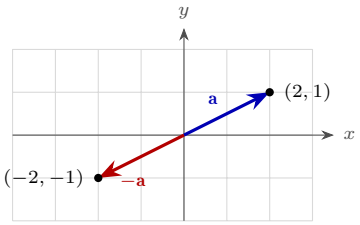
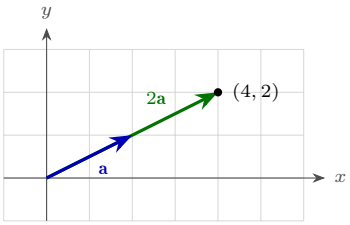
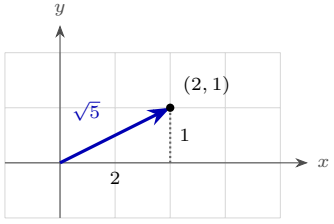
13.1 Vectors in Two Dimensions

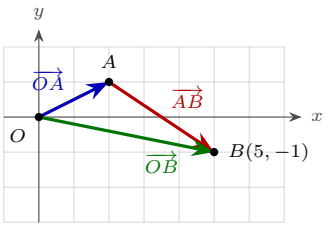
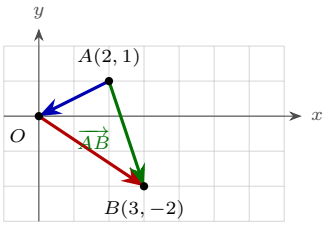
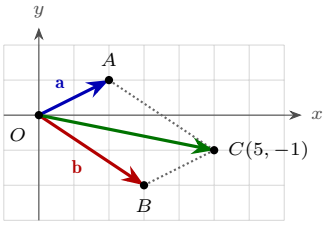
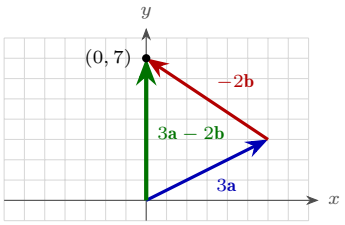
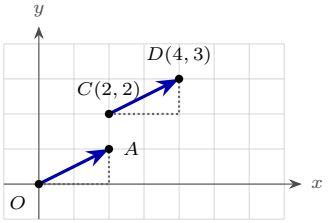
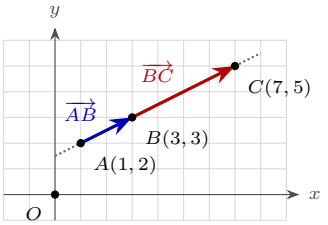
In Secondary 4, we have learned vectors in two dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$$

A vector describes a quantity that has both magnitude and direction. Vectors let us describe position, displacement and direction algebraically, so that geometric problems can be solved by calculation. In H2 Mathematics, we extend vectors from two dimensions to three dimensions, and use them to work with lines and planes, angles, distances, intersections and the foot of the perpendicular. Outside mathematics, vectors are used for forces and velocities in physics, in navigation, in computer graphics and 3D modelling, and throughout engineering. Vectors also lie at the heart of mathematics in linear algebra, which has foundational applications in fields such as physics, engineering, computer science, data science, machine learning and artificial intelligence, economics and cryptography.



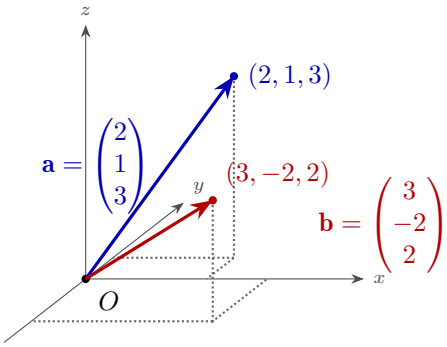
Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$	
$2\mathbf{a}$	$k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \end{pmatrix}\right\rvert = \sqrt{x^2 + y^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2 + 1^2}$ $= \sqrt{5}$	

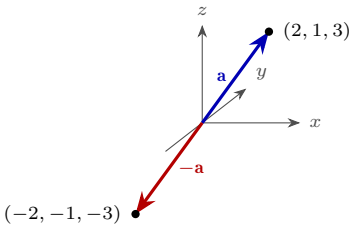
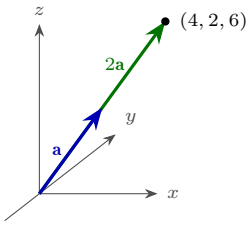
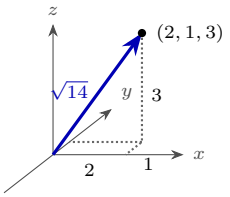
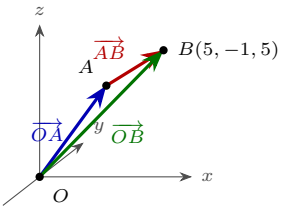
Name	Equation	Diagram
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 4-2 \\ 3-2 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 3-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 7-3 \\ 5-3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

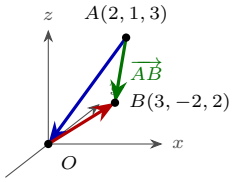
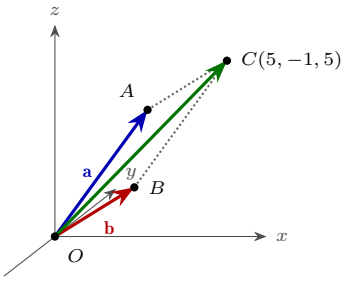
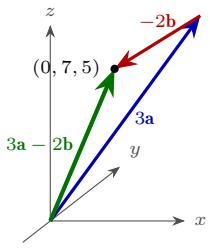
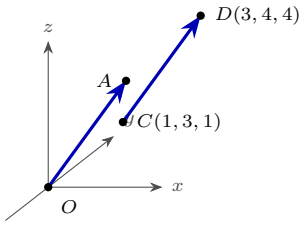
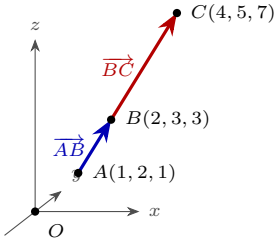
13.2 Vectors in Three Dimensions

In H2 Mathematics, we extend vectors to three dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$



Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \\ -3 \end{pmatrix}$	
2a	$k\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} kx \\ ky \\ kz \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\rvert = \sqrt{x^2+y^2+z^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2+1^2+3^2}$ $= \sqrt{14}$	
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	

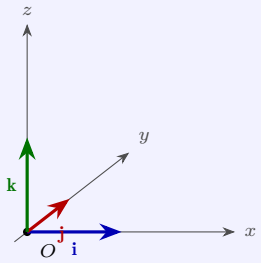
Name	Equation	Diagram
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \\ \lambda a_3 + \mu b_3 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 3-1 \\ 4-3 \\ 4-1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 2-1 \\ 3-2 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 4-2 \\ 5-3 \\ 7-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

13.3 Basic Terminology and Notation

Definition 13.3.1. Unit vectors i, j, k

i, **j** and **k** are the unit vectors in the positive directions of the x-, y- and z-axes:

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



Any vector can be written in terms of **i**, **j**, **k**:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Example 13.3.1

Write in column vector form: (a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ (b) $3\mathbf{i} + 2\mathbf{k}$. (c) Write $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$ in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Solution.

(a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$ (b) $3\mathbf{i} + 2\mathbf{k} = 3\mathbf{i} + 0\mathbf{j} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$ (c) $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix} = 5\mathbf{i} - \mathbf{j}$

Theorem 13.3.1. Magnitude of a vector

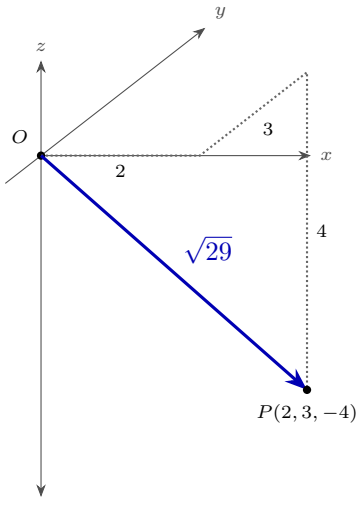
$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad \Rightarrow \quad |\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

Example 13.3.2

Find $|2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}|$.

Solution.

$$\begin{aligned} |2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}| &= \sqrt{2^2 + 3^2 + (-4)^2} \\ &= \sqrt{4 + 9 + 16} \\ &= \sqrt{29} \end{aligned}$$



Theorem 13.3.2. Parallel vectors

$$\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} = \lambda\mathbf{b} \text{ for some scalar } \lambda \neq 0 \quad (\text{alternatively, } \mathbf{a} \times \mathbf{b} = \mathbf{0})$$

$\lambda > 0$: same direction; $\lambda < 0$: opposite directions.

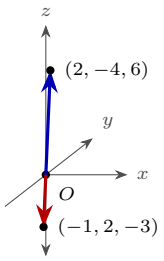
Example 13.3.3

(a) Show that $\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$. (b) Find p and q such that $\begin{pmatrix} p \\ 6 \\ q \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$.

Solution.

(a) $$\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} \parallel \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} \text{ (shown)}$$



(b) $$\begin{pmatrix} p \\ 6 \\ q \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

$$6 = 3\lambda \Rightarrow \lambda = 2$$

$$\therefore p = 2, \quad q = -4$$

Theorem 13.3.3. Unit vectors

The unit vector in the direction of $\mathbf{a}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Example 13.3.4

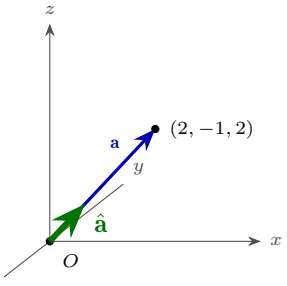
Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Find (a) the unit vector in the direction of $\mathbf{a}$, (b) the vectors of magnitude 6 parallel to $\mathbf{a}$.

Solution.

(a) $$|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$$

$$\hat{\mathbf{a}} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

(b) $$\pm 6\hat{\mathbf{a}} = \pm \frac{6}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \pm \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix}$$



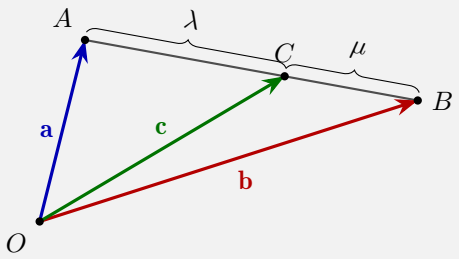
13.3.1 Ratio Theorem

Theorem 13.3.4. Ratio Theorem

If C lies on AB with $AC : CB = \lambda : \mu$, then

$$\overrightarrow{OC} = \frac{\mu\overrightarrow{OA} + \lambda\overrightarrow{OB}}{\lambda + \mu}$$

i.e. $\mathbf{c} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$.



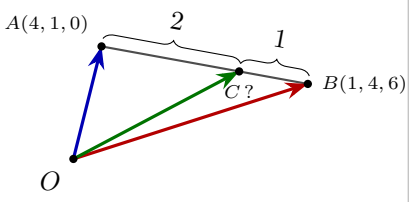
Example 13.3.5

$\overrightarrow{OA} = \begin{pmatrix}4\\1\\0\end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix}1\\4\\6\end{pmatrix}$. C lies on AB with $AC : CB = 2 : 1$. Find $\overrightarrow{OC}$.

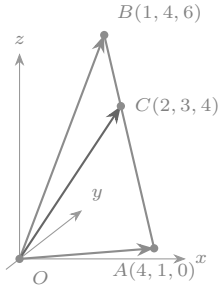
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{1\overrightarrow{OA} + 2\overrightarrow{OB}}{2+1}\\ &= \frac{1}{3}\left[\begin{pmatrix}4\\1\\0\end{pmatrix} + \begin{pmatrix}2\\8\\12\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}6\\9\\12\end{pmatrix}\\ &= \begin{pmatrix}2\\3\\4\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



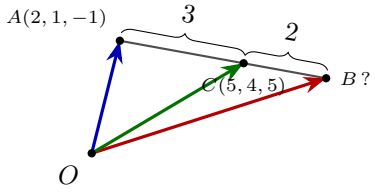
Example 13.3.6

A has position vector $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and C has position vector $\begin{pmatrix}5\\4\\5\end{pmatrix}$. C lies on AB with $AC : CB = 3 : 2$. Find $\overrightarrow{OB}$.

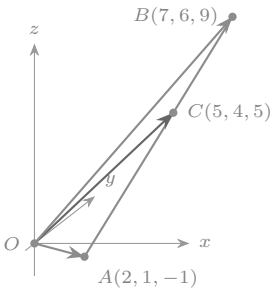
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{2\overrightarrow{OA} + 3\overrightarrow{OB}}{3+2}\\ \therefore\ \overrightarrow{OB} &= \frac{5\overrightarrow{OC} - 2\overrightarrow{OA}}{3}\\ &= \frac{1}{3}\left[\begin{pmatrix}25\\20\\25\end{pmatrix} - \begin{pmatrix}4\\2\\-2\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}21\\18\\27\end{pmatrix}\\ &= \begin{pmatrix}7\\6\\9\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



13.4 Scalar Product

Theorem 13.4.1. Scalar product

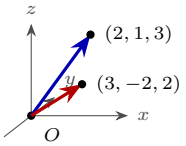
$$\mathbf{a}\cdot\mathbf{b}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}\cdot\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}=a_1b_1+a_2b_2+a_3b_3$$

Example 13.4.1

$\mathbf{a}=\begin{pmatrix}2\\1\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}3\\-2\\2\end{pmatrix}$. Find $\mathbf{a}\cdot\mathbf{b}$.

Solution.

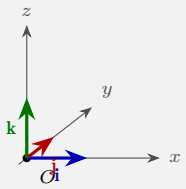
$$\begin{aligned}\mathbf{a}\cdot\mathbf{b}&=2(3)+1(-2)+3(2)\\&=6-2+6\\&=10\end{aligned}$$



Theorem 13.4.2. Properties

$$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}\qquad \mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$$
$$\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}\qquad \mathbf{i}\cdot\mathbf{i}=\mathbf{j}\cdot\mathbf{j}=\mathbf{k}\cdot\mathbf{k}=1$$
$$\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})\qquad \mathbf{i}\cdot\mathbf{j}=\mathbf{j}\cdot\mathbf{k}=\mathbf{k}\cdot\mathbf{i}=0$$

$\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law



Example 13.4.2

(a) $\mathbf{a}=\begin{pmatrix}1\\2\\-2\end{pmatrix}$. Show that $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$. (b) $|\mathbf{a}|=5$, $|\mathbf{b}|=3$. Find $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})$. (c) $\mathbf{a}\cdot\mathbf{b}=10$. Find $(2\mathbf{a})\cdot(3\mathbf{b})$. (d) Find $\mathbf{i}\cdot(2\mathbf{i}+3\mathbf{j}-\mathbf{k})$. (e) $\mathbf{p}=\mathbf{i}$, $\mathbf{q}=\mathbf{i}+2\mathbf{j}$, $\mathbf{r}=\mathbf{i}+5\mathbf{k}$. Show that $\mathbf{p}\cdot\mathbf{q}=\mathbf{p}\cdot\mathbf{r}$ but $\mathbf{q}\neq\mathbf{r}$, and explain why $\mathbf{p}\cdot\mathbf{q}\cdot\mathbf{r}$ is meaningless.

Solution.

(a) $\mathbf{a}\cdot\mathbf{a}=1^2+2^2+(-2)^2=9=|\mathbf{a}|^2$

$\therefore\ \mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$ (shown)

(b) $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\mathbf{a}\cdot\mathbf{a}-\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b}$

$=|\mathbf{a}|^2-|\mathbf{b}|^2$

$=25-9=16$

(c) $(2\mathbf{a})\cdot(3\mathbf{b})=6(\mathbf{a}\cdot\mathbf{b})$

$=60$

(d) $\mathbf{i}\cdot(2\mathbf{i}+3\mathbf{j}-\mathbf{k})=2\,\mathbf{i}\cdot\mathbf{i}+3\,\mathbf{i}\cdot\mathbf{j}-\mathbf{i}\cdot\mathbf{k}$

$=2(1)+3(0)-0$

$=2$

(e) $\mathbf{p}\cdot\mathbf{q}=1+0+0=1$

$\mathbf{p}\cdot\mathbf{r}=1+0+0=1$

$\therefore\ \mathbf{p}\cdot\mathbf{q}=\mathbf{p}\cdot\mathbf{r},\ \mathbf{q}\neq\mathbf{r}$ (shown)

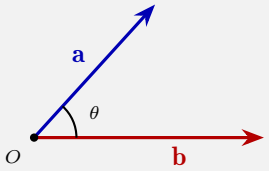
$\mathbf{p}\cdot\mathbf{q}=1$ (scalar)

$\therefore\ (\mathbf{p}\cdot\mathbf{q})\cdot\mathbf{r}=1\cdot\mathbf{r}$ undefined

Theorem 13.4.3. Formula and angle

$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$
$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

$\mathbf{a}\cdot\mathbf{b} > 0$ acute, < 0 obtuse, $= 0$ right angle
θ: tail to tail, $0 \le \theta \le \pi$



Example 13.4.3

(a) $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$, angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{3}$. Find $\mathbf{a}\cdot\mathbf{b}$. (b) Find the angle between $\begin{pmatrix}2\\1\\3\end{pmatrix}$ and $\begin{pmatrix}3\\-2\\2\end{pmatrix}$. (c) Find the angle between $\mathbf{i}+2\mathbf{j}-2\mathbf{k}$ and $2\mathbf{i}-\mathbf{j}+3\mathbf{k}$.

Solution.

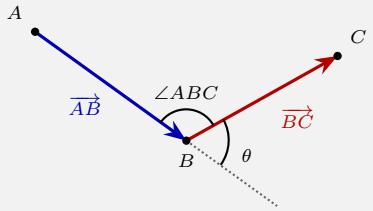
(a) $\mathbf{a}\cdot\mathbf{b} = 3(4)\cos\frac{\pi}{3}$
$= 6$

(b) $\cos\theta = \frac{6-2+6}{\sqrt{14}\sqrt{17}}$
$= \frac{10}{\sqrt{238}}$ $(> 0,$ acute$)$
$\theta = 49.6°$ (1 d.p.)

(c) $\cos\theta = \frac{2-2-6}{\sqrt{1^2+2^2+(-2)^2}\sqrt{2^2+(-1)^2+3^2}}$
$= \frac{-6}{3\sqrt{14}}$ $(< 0,$ obtuse$)$
$\theta = 122.3°$ (1 d.p.)

Theorem 13.4.4. Angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$

$\angle ABC \neq$ angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$
$$\cos\angle ABC = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|}$$



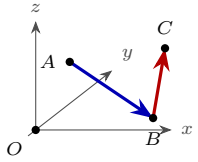
Example 13.4.4

$A(1,0,2)$, $B(3,1,0)$, $C(2,4,1)$. Find (a) $\angle ABC$, (b) the angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$.

Solution.

(a) $\overrightarrow{BA} = \begin{pmatrix}-2\\-1\\2\end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix}-1\\3\\1\end{pmatrix}$
$\cos\angle ABC = \frac{2-3+2}{3\sqrt{11}}$
$\angle ABC = 84.2°$ (1 d.p.)

(b) $\overrightarrow{AB} = \begin{pmatrix}2\\1\\-2\end{pmatrix}$
$\cos\theta = \frac{-2+3-2}{3\sqrt{11}}$
$\theta = 95.8°$ (1 d.p.)

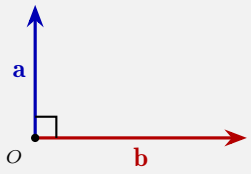


Theorem 13.4.5. Perpendicular vectors

$$\mathbf{a}\cdot\mathbf{b}=0 \iff \mathbf{a}\perp\mathbf{b}$$

($\mathbf{a}$, $\mathbf{b}$ nonzero)

$\because \cos\dfrac{\pi}{2}=0$



Example 13.4.5

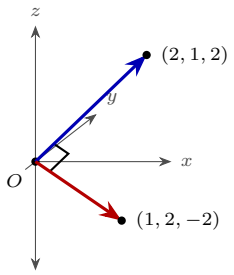
(a) Show that $\begin{pmatrix}2\\1\\2\end{pmatrix}$ is perpendicular to $\begin{pmatrix}1\\2\\-2\end{pmatrix}$. (b) Find p such that $\begin{pmatrix}2\\p\\-1\end{pmatrix}$ is perpendicular to $\begin{pmatrix}3\\1\\4\end{pmatrix}$.

Solution.

(a) $\begin{pmatrix}2\\1\\2\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-2\end{pmatrix}=2+2-4$

$=0$

$\therefore \begin{pmatrix}2\\1\\2\end{pmatrix}\perp\begin{pmatrix}1\\2\\-2\end{pmatrix}$ (shown)



(b) $6+p-4=0$

$\therefore\ p=-2$

Example 13.4.6

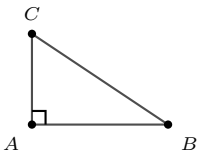
$A(1,1,1)$, $B(3,2,3)$, $C(2,3,-1)$. Show that triangle ABC is right-angled at A.

Solution.

$$\overrightarrow{AB}=\begin{pmatrix}2\\1\\2\end{pmatrix},\quad \overrightarrow{AC}=\begin{pmatrix}1\\2\\-2\end{pmatrix}$$

$\overrightarrow{AB}\cdot\overrightarrow{AC}=2+2-4=0$

$\therefore\ \angle BAC=90^\circ$ (shown)



Example 13.4.7

(a) Find a vector perpendicular to both $\mathbf{i}+\mathbf{j}$ and $\mathbf{j}+\mathbf{k}$. (b) $|\mathbf{a}+\mathbf{b}|=|\mathbf{a}-\mathbf{b}|$, $\mathbf{a},\mathbf{b}$ nonzero. Show that $\mathbf{a}\perp\mathbf{b}$.

Solution.

(a) $x+y=0,\quad y+z=0$

$x=1\ \Rightarrow\ y=-1,\ z=1$

$\therefore\ \mathbf{i}-\mathbf{j}+\mathbf{k}$

(b) $|\mathbf{a}+\mathbf{b}|^2=|\mathbf{a}-\mathbf{b}|^2$

$|\mathbf{a}|^2+2\,\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2=|\mathbf{a}|^2-2\,\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2$

$4\,\mathbf{a}\cdot\mathbf{b}=0$

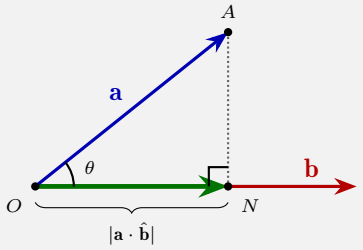
$\therefore\ \mathbf{a}\perp\mathbf{b}$ (shown)

Theorem 13.4.6. Applications: projection of a onto b

Length of projection $= |\mathbf{a}\cdot\hat{\mathbf{b}}|$

Projection vector $= (\mathbf{a}\cdot\hat{\mathbf{b}})\,\hat{\mathbf{b}}$

$|\mathbf{a}\cdot\hat{\mathbf{b}}| = ON$, $\overrightarrow{ON} = (\mathbf{a}\cdot\hat{\mathbf{b}})\,\hat{\mathbf{b}}$



Example 13.4.8

$\mathbf{a} = \begin{pmatrix}3\\3\\3\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}1\\2\\2\end{pmatrix}$. Find (a) the length of projection of $\mathbf{a}$ onto $\mathbf{b}$, (b) the projection vector of $\mathbf{a}$ onto $\mathbf{b}$, (c) the perpendicular distance from A to the line through O parallel to $\mathbf{b}$.

Solution.

(a) $\hat{\mathbf{b}} = \frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$

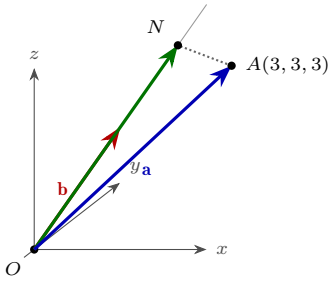
$$|\mathbf{a}\cdot\hat{\mathbf{b}}| = \frac{|3+6+6|}{3} = 5$$

(b) $\overrightarrow{ON} = 5\left(\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right) = \frac{5}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$

(c) $AN^2 = |\mathbf{a}|^2 - ON^2$

$= 27 - 25$

$AN = \sqrt{2}$



Example 13.4.9

$\mathbf{a} = 2\mathbf{i}-\mathbf{j}+3\mathbf{k}$, $\mathbf{d} = \mathbf{i}+\mathbf{j}+\mathbf{k}$. Find (a) the length of projection of $\mathbf{a}$ onto $\mathbf{d}$, (b) the projection vector of $\mathbf{a}$ onto $\mathbf{d}$, (c) the perpendicular distance from A to the line through O parallel to $\mathbf{d}$.

Solution.

(a) $\hat{\mathbf{d}} = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}$

$$|\mathbf{a}\cdot\hat{\mathbf{d}}| = \frac{|2-1+3|}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

(b) $(\mathbf{a}\cdot\hat{\mathbf{d}})\,\hat{\mathbf{d}} = \frac{4}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}$

$= \frac{4}{3}\begin{pmatrix}1\\1\\1\end{pmatrix}$

(c) $d^2 = |\mathbf{a}|^2 - |\mathbf{a}\cdot\hat{\mathbf{d}}|^2$

$= 14 - \frac{16}{3}$

$d = \sqrt{\frac{26}{3}}$